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Dept: Data Engineering

Module: ADF: Key Features, Use Cases, Key Terms

Task:

Create Linked Services to connect Azure Data Factory with Azure Data Lake Storage (ADLS) and Microsoft SQL Server database. Configure necessary authentication details (e.g., Azure Key Vault for secrets).

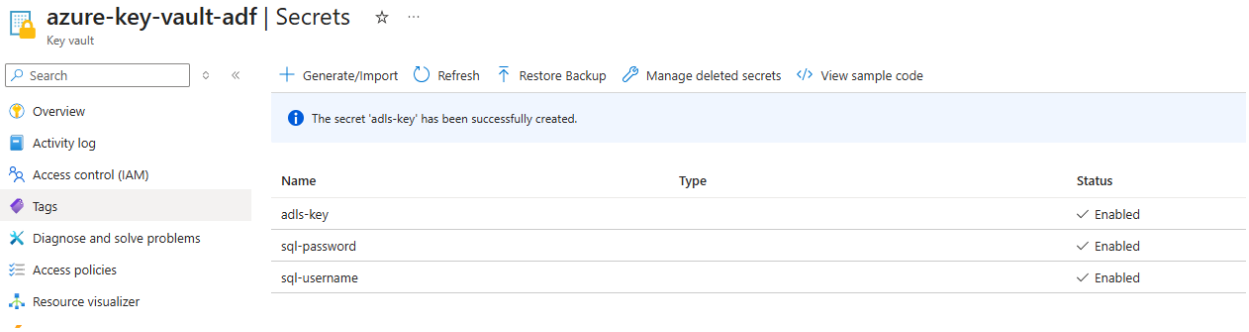
Create datasets for both the source (CSV files in ADLS) and the destination (tables in SQL Server).

Initially create the following resources in Azure:

ADF, SQL database and blob container (in storage account)

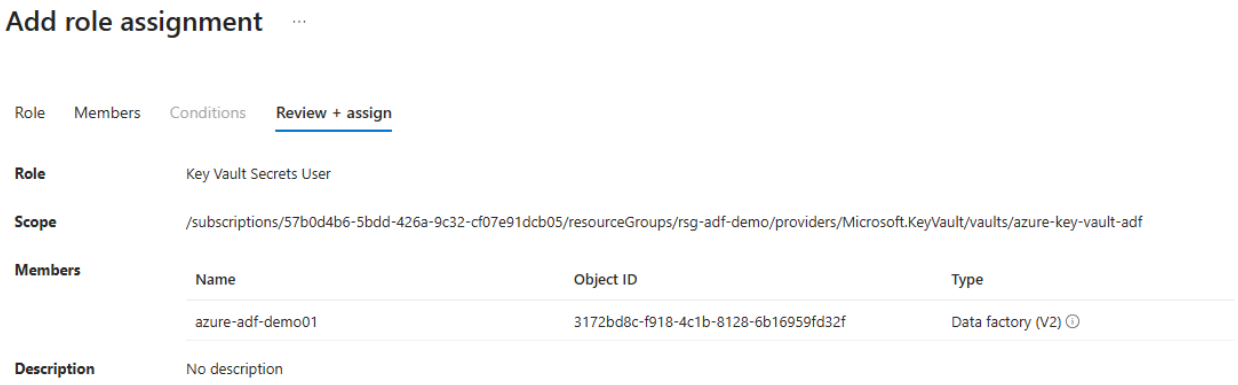
Create key-vault and store necessary credentials

- Storage account connection string → `adls-key`
- SQL username → `sql-username`
- SQL password → `sql-password`



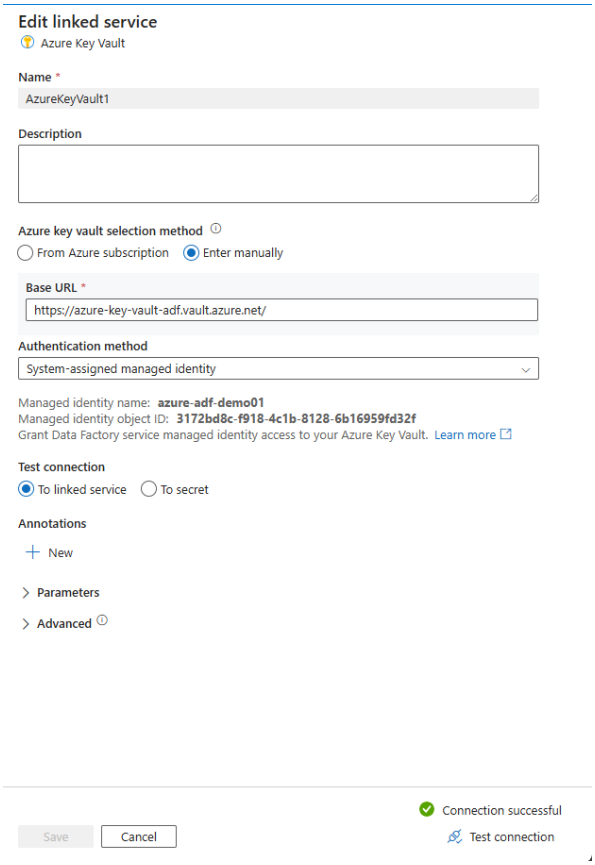
Add role assignment to ADF (as managed identity) from key-vault so that it can access the secrets

Make sure that you store connection-string in adls-key instead of access-key



Add linked services for blob and sql database:

And for secure credential management, create linked service for azure key vault that connects with key-vault and use the secrets



Edit linked service

Azure Blob Storage Learn more

Name *

inputBlob

Description

Connect via integration runtime *

AutoResolveIntegrationRuntime

Authentication type

Account key

Connection string | Azure Key Vault

AKV linked service *

AzureKeyVault1

Secret name *

adls-key

Edit

Secret version

Latest version

Edit

Test connection

To linked service | To file path

Annotations

New

Parameters

New | Delete

Connection successful

Test connection

Save | Cancel

Edit linked service

Azure SQL Database Learn more

Name *
outputSQL

Description

Connect via integration runtime *
AutoResolveIntegrationRuntime

Version
2.0 (Recommended) 1.0

Account selection method
From Azure subscription Enter manually

Fully qualified domain name *
azure-sql-server-adf.database.windows.net

Database name *
azure-sql-db-adf

Authentication type *
SQL authentication

User name *
sqladmin

Password Azure Key Vault

AKV linked service *
AzureKeyVault1

Secret name *
sql-password

Edit

Connection successful

Save Cancel Test connection

Linked services

Linked service defines the connection information to a data store or compute. Learn more

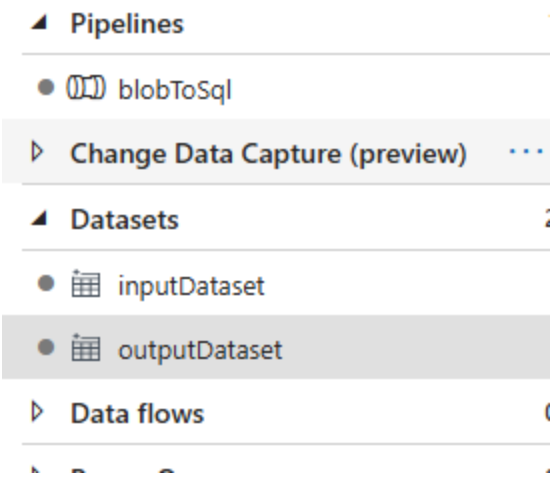
+ New

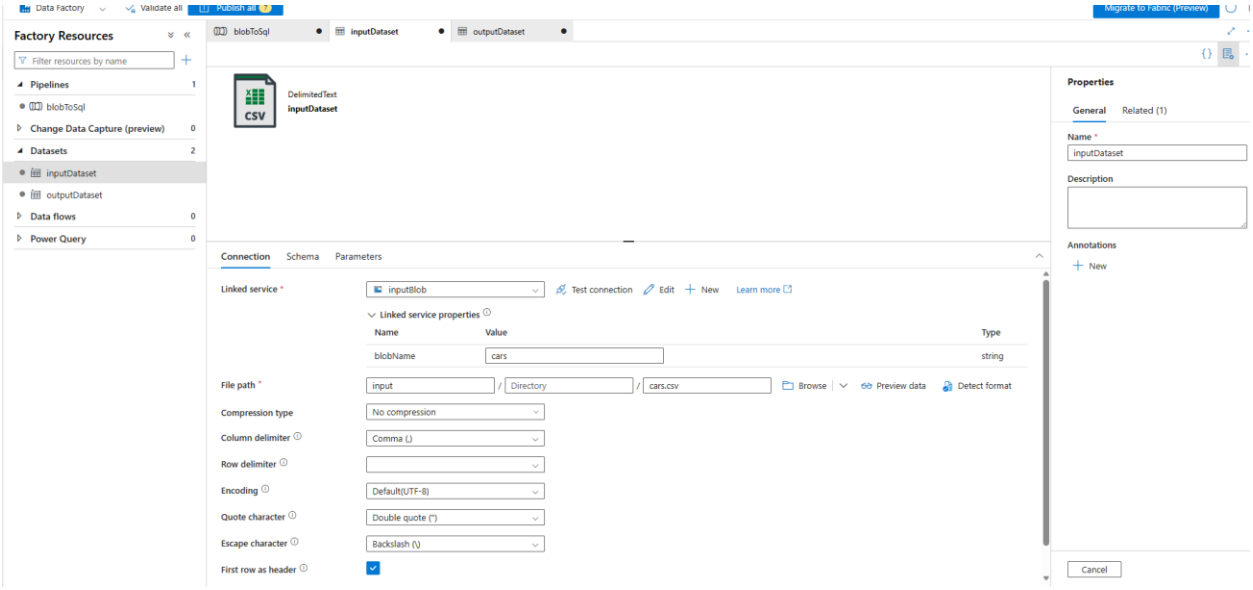
Filter by name Annotations : Any

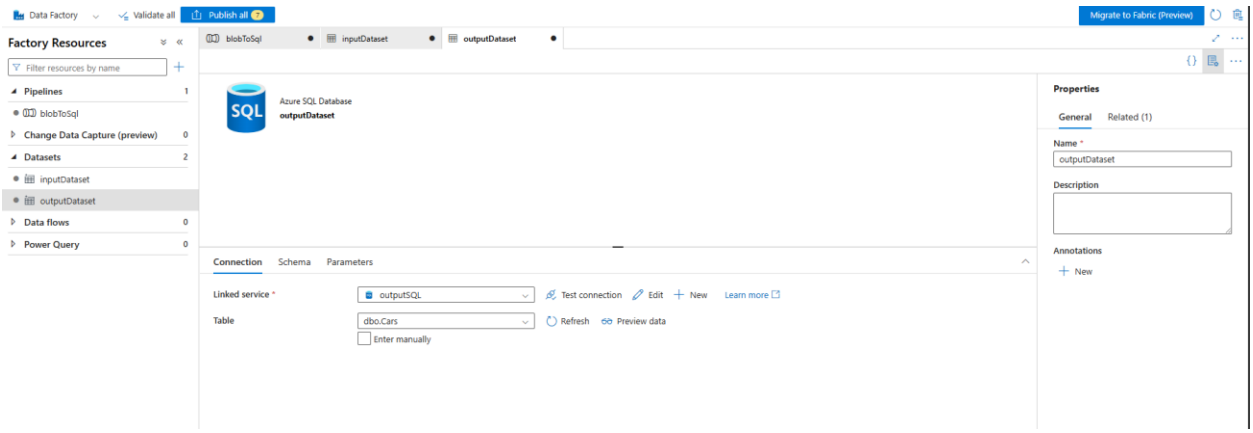
Showing 1 - 3 of 3 items

Name	Type	Related
AzureKeyVault1	Azure Key Vault	2
inputBlob	Azure Blob Storage	1
outputSQL	Azure SQL Database	1

Then add the datasets in pipeline


Pipelines
blobToSql
Change Data Capture (preview)
Datasets
inputDataset
outputDataset
Data flows


Data Factory
Validate all
Publish all
Migrate to Fabric (Preview)
Factory Resources
Filter resources by name
Pipelines 1
blobToSql
Change Data Capture (preview) 0
Datasets 2
inputDataset
outputDataset
Data flows 0
Power Query 0
blobToSql
inputDataset
outputDataset
DelimitedText
inputDataset
Properties
General
Related (1)
Name *
inputDataset
Description
Annotations
New
Connection
Schema
Parameters
Linked service *
inputBlob
Test connection
Edit
New
Learn more
Linked service properties
Name
Value
Type
blobName
cars
string
File path *
input
Directory
cars.csv
Browse
Preview data
Detect format
Compression type
No compression
Column delimiter
Comma (,)
Row delimiter
Encoding
Default(UTF-8)
Quote character
Double quote (")
Escape character
Backslash (\)
First row as header
Cancel


Data Factory
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Migrate to Fabric (Preview)
Factory Resources
Filter resources by name
Pipelines 1
blobToSql
Change Data Capture (preview) 0
Datasets 2
inputDataset
outputDataset
Data flows 0
Power Query 0
blobToSql
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outputDataset
Azure SQL Database
outputDataset
Properties
General
Related (1)
Name *
outputDataset
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Linked service *
outputSQL
Test connection
Edit
New
Learn more
Table
dbo.Cars
Refresh
Preview data
Enter manually